

We can solve the following state equation using the ode23 command.

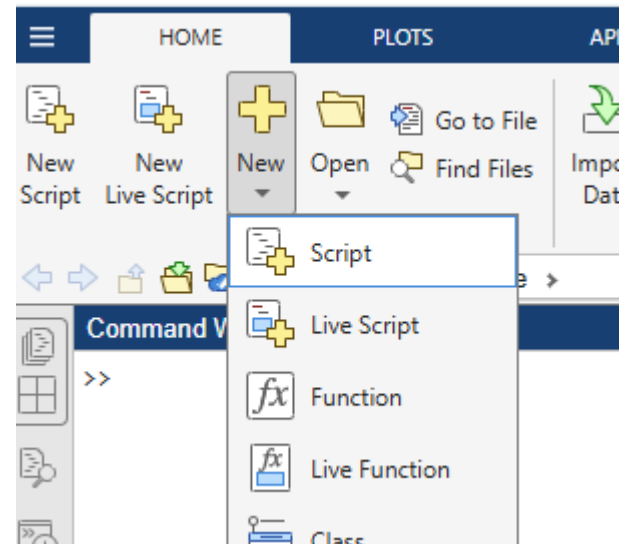
u = unit step

Here we assume initial conditions are zero as in the homework problem.

$$\dot{x}_1 = -4x_1 - 4x_2 + 4u$$

$$\dot{x}_2 = -.25x_1$$

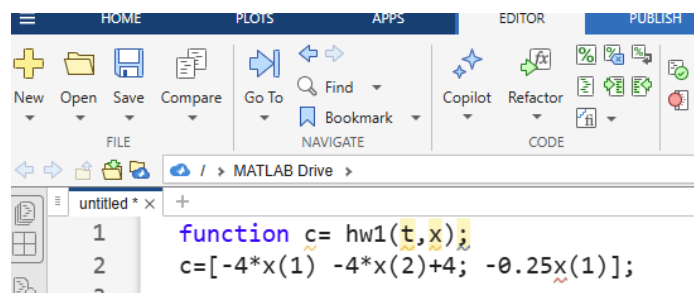
1. Open MATLAB. Open a new script window.



2. You are going to create a function named **hw1**

Type the following in the script window

```
function c= hw1(t,x);  
c=[-4*x(1)-4*x(2)+4; -0.25*x(1)];
```



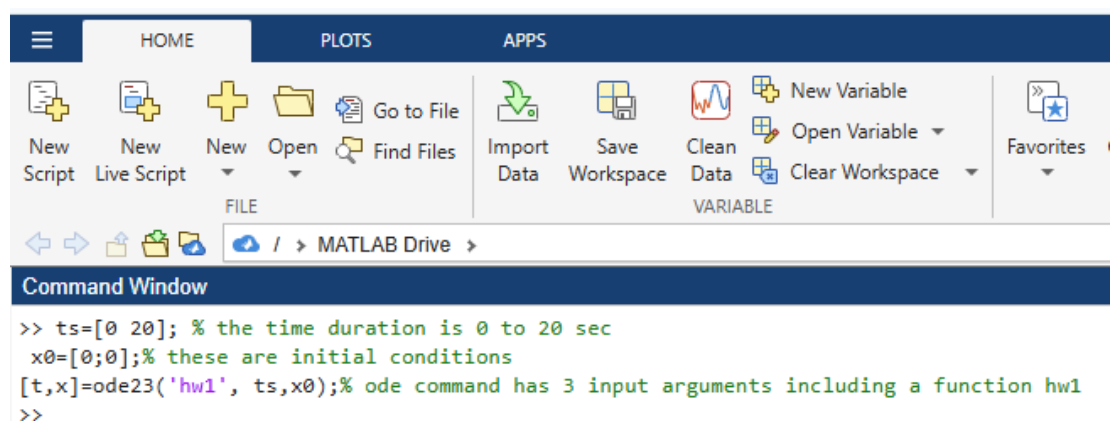
Save the file as .m file in MATLAB drive. When you save it, automatically the file name become hw1.m

3. After closing the script window, write the following in the command window and press **Enter**.

```
ts=[0 20]; % the time duration is 0 to 20 sec
```

```
x0=[0;0];% these are initial conditions
```

```
[t,x]=ode23('hw1', ts,x0);% ode command has 3 input arguments
```



The ode command's outputs are **t** and **x**

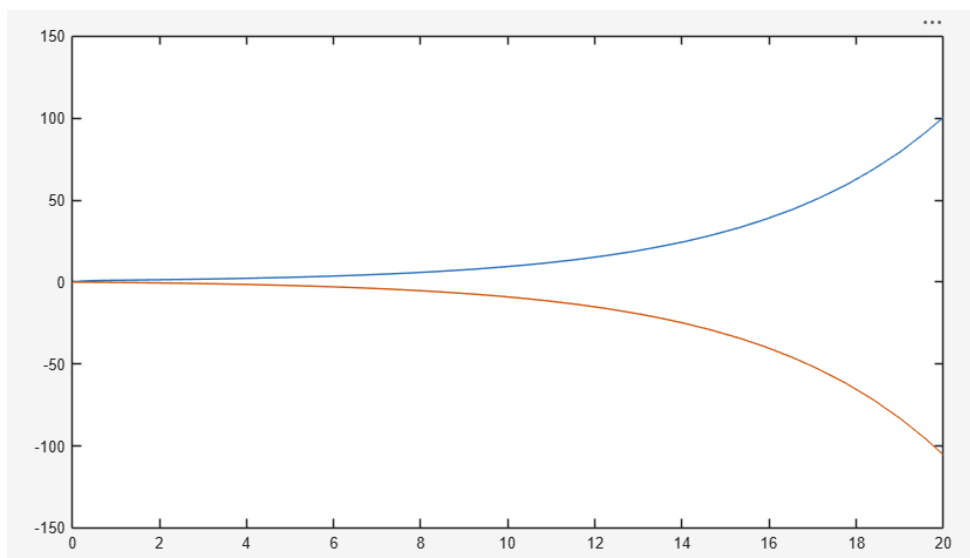
Just write **t** in command window and press Enter. It will display time 0 to 20 in a column,

Then write **size(x)** and press Enter. Then you can write simply **x** and press Enter.

x is the state consisting of two columns **x1** and **x2**

If you write **[t,x]** in command window, what you will get?

4. Write **plot(t,x)** in the command window and press Enter, you will be getting the time variation of state variables **x1** and **x2** in the same plot



5. When you write **plot(x(:,1),x(:,2))** you will be getting a state trajectory.